



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
28-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TAAS, TTAS & TEAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

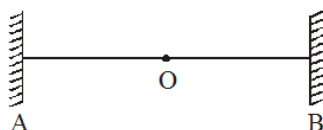
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

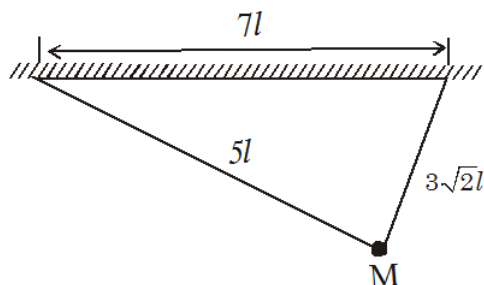
1. An infinite solenoid has radius R and n turns per unit length. The current grows linearly with time, according to $I_t = Ct$, in the solenoid. Here C is some constant. Let the induced electric field at distance r from axis of solenoid is E . Choose correct alternative(s).
 - (A) $E \propto r$ for $r < R$
 - (B) $E \propto 1/r$ for $r > R$
 - (C) If an infinite line charge having uniform linear charge density λ is placed along the axis, then electrostatic field produced by line charge and induced electric field are perpendicular to each other
 - (D) The induced field and electrostatic field produced by line charge placed along axis of solenoid can be added vectorically to get net electric field at a point.

Space for Rough Work

2. A spring of spring constant K is fixed to the ceiling of a lift. The other end of the spring is attached to a block of mass m . The mass is in equilibrium. Now the lift accelerates downwards with an acceleration $2g$.
- (A) Block will not perform SHM and it will touch the ceiling and remain stuck with it.
- (B) Block will perform SHM of time period $2\pi\sqrt{m/K}$
- (C) Amplitude of SHM of the block is $2mg/K$
- (D) Minimum potential energy of spring during motion is zero
3. A wire under tension between two fixed point A & B, executes sustained transverse vibrations so that the midpoint O of AB is a NODE, then



- (A) There are no two points on the string which have a phase difference of $\frac{\pi}{8}$.
- (B) Total mechanical energy of a small element of string at O, remain constant.
- (C) Total mechanical energy of the string AB remain constant.
- (D) There can be only two points between A & O, which will have equal speeds at all times.
4. A particle is suspended by two ideal strings as shown in the figure. Now mass M is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).



- (A) The period of small oscillation of the system is $2\pi\sqrt{\frac{3\sqrt{3}l}{g}}$.
- (B) The period of small oscillation of the system is $2\pi\sqrt{\frac{3l}{g}}$
- (C) The period of small oscillation of the system is independent of M .
- (D) The period of small oscillation increases with M .

Space for Rough Work

5. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency $4 \times 10^5 \text{ rads}^{-1}$. At resonance the voltages across resistance and inductance are 60 V and 40V respectively.

(a) The value of L and C are 0.2 mH, $\frac{1}{32}\mu\text{F}$

(b) $8 \times 10^5 \text{ rad/s}$, the current lags the voltage by 45°

(c) $6 \times 10^5 \text{ rad/s}$, the current lags the voltage by 45°

(A) (a), (c) are correct

(B) (a), (b) are correct

(C) (a), (b), (c) are correct

(D) (a), (b), (c) are wrong

6. Two pulses travelling on same string are described by equations

$$y_1 = \frac{5}{(3x - 4t)^2 + 2} \text{ and } y_2 = \frac{-5}{(3x + 4t - 6)^2 + 2}$$

Mark out correct statements

(A) At $t = \frac{4}{3}\text{s}$ the two waves cancel each other everywhere

(B) At $t = \frac{3}{4}\text{s}$ the two waves cancel each other everywhere

(C) At $x = 1$ the two waves cancel each other at all times

(D) At all points two waves cancel each other

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. In the equation $X = A \sin (\omega t + \pi/4)$ match the following :

List-I		List-II	
(P)	Kinetic energy at $x = A/2$	(1)	Zero
(Q)	Potential energy at $x = A/2$	(2)	Maximum
(R)	Kinetic energy at $x = A$	(3)	$\frac{1}{4}$ times the maximum value
(S)	Potential energy at $x = A$	(4)	$\frac{3}{4}$ times the maximum value

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

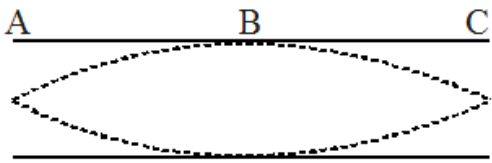
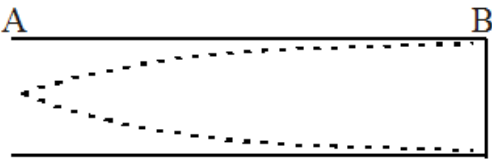
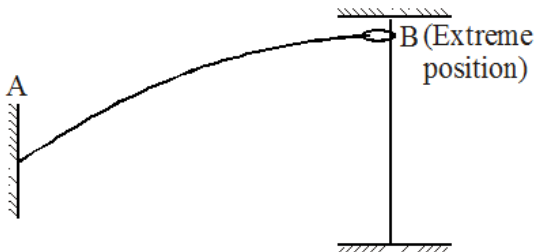
(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 1$

Space for Rough Work

8.

List-I		List-II	
(P)	Graphical representation of pressure variation in both end open organ pipe. 	(1)	Maximum kinetic energy at B
(Q)	Graphical representation of pressure variation in one end closed organ pipe. 	(2)	Maximum potential energy at B
(R)	Snapshot of string fixed at both ends. 	(3)	Maximum particle velocity at B
(S)	Snap shot of a string fixed at one end and connected to a smooth massless ring that is constrained to move vertically. 	(4)	Maximum particle acceleration at B

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 2; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

Space for Rough Work

9. A charged particle with some initial velocity is projected in a region, where non-zero, non-varying and uniform electric and / or magnetic fields are present. In List-I, information about existence of electric field and / or magnetic field, direction of initial velocity of charged particle are given while in List-II, the probable path of charged particle is mentioned. Match the entries of List-I with entries of List-II: (consider gravity free space)

List-I		List-II	
(P)	$\vec{E} = 0, \vec{B} \neq 0$ and initial velocity is at an unknown angle with $\vec{B}$	(1)	Straight line
(Q)	$\vec{E} \neq 0, \vec{B} = 0$ and initial velocity is at unknown angle with $\vec{E}$	(2)	Parabola
(R)	$\vec{E} \neq 0, \vec{B} \neq 0$, $\vec{E}$ parallel to $\vec{B}$ and initial velocity is perpendicular to $\vec{E}$	(3)	Circular
(S)	$\vec{E} \neq 0, \vec{B} \neq 0$, $\vec{E}$ perpendicular to $\vec{B}$ and initial velocity is perpendicular to both $\vec{E}$ and $\vec{B}$	(4)	Helical path with non-uniform pitch
		(5)	Cycloid

(A) $P \rightarrow 1,3; Q \rightarrow 1,2; R \rightarrow 4; S \rightarrow 1,5$

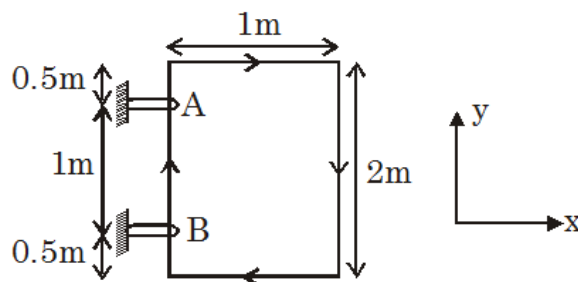
(B) $P \rightarrow 2,3; Q \rightarrow 1,4,5; R \rightarrow 1,3; S \rightarrow 4$

(C) $P \rightarrow 1,3; Q \rightarrow 1,2; R \rightarrow 1,4; S \rightarrow 2,4$

(D) $P \rightarrow 2,5; Q \rightarrow 3,4; R \rightarrow 4; S \rightarrow 1,5$

Space for Rough Work

10. A rigid uniform wire frame of mass 1 kg has dimension as shown. It is free to rotate about vertical axis supported by two hinges as shown. A current of 1A flows in the frame. A magnetic field $\vec{B}$ is on. Assume that the current in the frame remains unchanged and frame is initially at rest.



List-I		List-II	
(P)	$\vec{B} = 4\hat{i}$ tesla	(1)	The frame starts rotating in anticlockwise direction as seen from above (along y-axis).
(Q)	$\vec{B} = -4\hat{i}$ tesla	(2)	The frame starts rotating in clockwise direction as seen from above (along y-axis).
(R)	$\vec{B} = 3\hat{j}$ tesla	(3)	Hinge A exerts a non zero force in +ve z direction.
(S)	$\vec{B} = 6\hat{k}$ tesla	(4)	Hinge B exerts a non zero force in +ve z direction.
		(5)	Hinge A exerts a non zero force in +ve x direction.
		(6)	Hinge B exerts a non zero force in +ve x direction.

- (A) $P \rightarrow 2,3,4,6; Q \rightarrow 1,5; R \rightarrow 4,6; S \rightarrow 6$ (B) $P \rightarrow 1,3,4; Q \rightarrow 2,5; R \rightarrow 2,4,6; S \rightarrow 1,6$
 (C) $P \rightarrow 2,3,4; Q \rightarrow 1,3,6; R \rightarrow 1,4,6; S \rightarrow 2,6$ (D) $P \rightarrow 1,3; Q \rightarrow 2,5,6; R \rightarrow 4,6; S \rightarrow 1,3,5$

Space for Rough Work

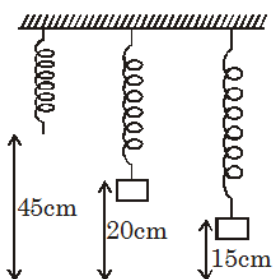
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

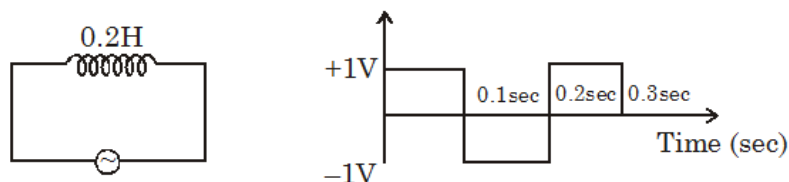
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A massless spring is suspended from a hook at the top. A small mass of 180 gm is suspended from the bottom of the hook. At the equilibrium it is at 20 cm from ground. It is pulled down to distance 15 cm from ground and released. Find the time (in sec) at which it is first at 25 cm from the ground. Take $\pi^2 = 10$.



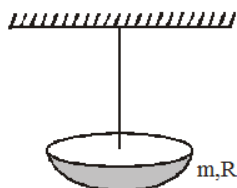
2. In a series LCR circuit, half power frequencies (frequency at which power dissipated is half the maximum power dissipated in the circuit) are found to be 90 rad/s and 40 rad/s. Find the value of $\frac{\omega_0}{10}$, where ω_0 is the resonance frequency.
3. An AC source has voltage given by



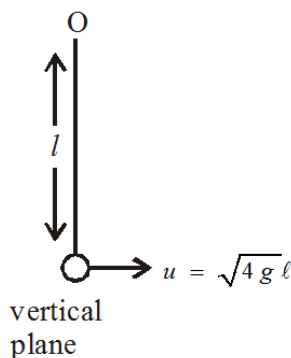
The circuit is connected at $t = 0$ sec. The current in the inductor (in ampere) at $t = 0.25$ sec is.

Space for Rough Work

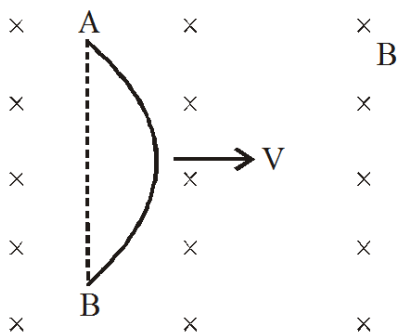
4. A uniform solid hemisphere of mass m and radius R is attached to the roof with a chord of torsional constant C and performing torsional SHM. Then the time period (in seconds) of SHM is (Take $m = 15 \text{ kg}$, $R = \frac{2}{\pi} \text{ m}$, $C = 6 \text{ Nm/rad}$).



5. A pendulum bob hinged at O is given velocity $u = \sqrt{4g\ell}$ from bottommost point as shown. If $\ell = 54 \text{ cm}$, find maximum height (in m) reached by bob from point of projection.

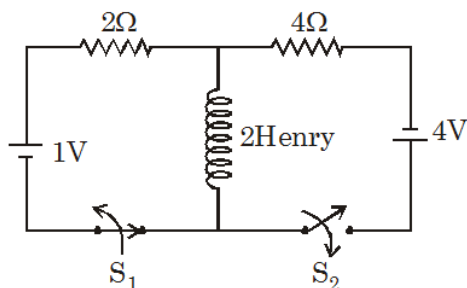


6. In the figure AB is a metallic wire bent in the form of a circular arc of radius 1 m and central angle 74° . If its velocity $V = 0.7 \text{ m/s}$ and magnetic field $B = 5 \text{ Tesla}$ then find induced emf in the wire. $\left[\sin 37^\circ = \frac{3}{5} \right]$



Space for Rough Work

7. In the circuit shown in the figure, S_1 remains closed for a long time and S_2 remains open. Now S_2 is closed and S_1 is opened. Find out the $\left| \frac{di}{dt} \right|$ (in Amp/sec) just after that moment :



8. A source of sound with natural frequency f_0 moves uniformly along a straight line. A stationary sound detector is located at a distance of 300 m from the straight line. When the source is closest to the detector, it records a frequency of $3f_0$ as received from the source. Find the distance of the source from the detector at the instant when it records natural frequency f_0 of the source. If your answer is $20(x)\sqrt{5(y)}$ where x & y are integers from 1 to 9 then fill value $(x) + (y)$.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Select correct statement(s) :

(A) Low pressure is favourable for evaporation of $\text{H}_2\text{O}(\ell)$

(B) The degree of dissociation of $\text{CaCO}_3(\text{s})$ decreases with increase in pressure

(C) If the equilibrium constant for $\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g})$ is 25, then equilibrium constant for $\text{AB}(\text{g}) \rightarrow \frac{1}{2}\text{A}_2(\text{g}) + \frac{1}{2}\text{B}_2(\text{g})$ is 0.2

(D) If solid product is added to an equilibrium mixture, the equilibrium will be unaffected.

2. Let the colour of the indicator HIn (colourless) will be visible only when its ionised form (pink) is 25% or more in a solution. Suppose HIn ($\text{pK}_a = 9.0$) is added to a solution of $\text{pH} = 9.6$. Predict what will happen (Take $\log 2 = 0.3$)

(A) pink colour will be visible

(B) pink colour will not be visible

(C) % of ionised form will be less than 25%

(D) % of ionised form will be more than 25%

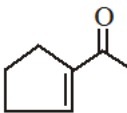
Space for Rough Work

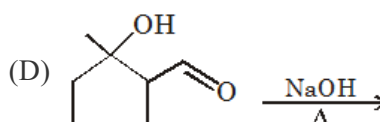
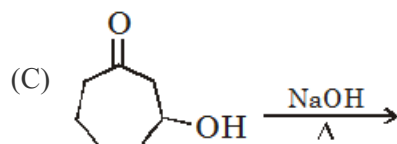
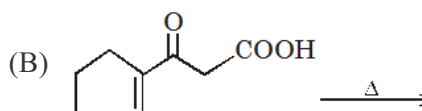
3. Potassium ferrocyanide is an important laboratory reagent used to detect many cations. Select the metal ions that can give coloured ppt. with it.

- (A) Zn^{2+} (B) Fe^{3+} (C) Cu^{2+} (D) Fe^{2+}

4. Which of the following are "True" statements?

- (A) Zn and Al salts can be distinguished by using NH_4OH .
 (B) Zn and Al salts can be distinguished by using NaOH
 (C) ZnS is precipitated by H_2S in ammonical solution of zinc salt but not in the acidic solution.
 (D) $\text{Al}(\text{OH})_3$ is soluble in NaOH as well as HCl

5. Which of the following reaction(s) can produce  as one of the product ?



6. Which of the following compound act as electrophile

- (A) CO_2 (B) LiAlH_4 (C) CH_3MgCl (D) KOH

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

List-I (Electrolysis using inert electrode)		List-II (Product at anode or cathode.)	
(P)	Aqueous solution of sodium acetate	(1)	H ₂
(Q)	Aqueous solution of sodium formate	(2)	CO ₂
(R)	Concentrated aqueous solution of NaCl	(3)	Cl ₂
(S)	Dilute aqueous solution of HCl	(4)	C ₂ H ₆
		(5)	O ₂

The correct option is :-

(A) P → 1,2,4; Q → 1,2; R → 1, 3; S → 1, 5

(B) P → 1,5; Q → 2,4; R → 1; S → 3, 4

(C) P → 1,4; Q → 2,4; R → 1, 5; S → 3, 5

(D) P → 1,2,4; Q → 1,4; R → 2, 5; S → 1,5

Space for Rough Work

8.

List-I (Electrolysis using inert electrode)		List-II (Product at anode or cathode.)	
(P)	$\text{CrO}_5 + \text{H}^+ \longrightarrow$	(1)	Redox reaction
(Q)	$\text{Cr}^{3+} + \text{K}_2\text{S}_2\text{O}_8 + \text{H}^+ \longrightarrow$	(2)	Evolution of O_2 is associated
(R)	$\text{MnO}_2 + \text{conc.KOH} + \text{O}_2 \xrightarrow{\Delta}$	(3)	Green colour solution/compound is formed as product
(S)	$\text{KMnO}_4 + \text{conc.KOH} \xrightarrow{\Delta}$	(4)	Orange/Yellow colour solution/compound is formed as product

The correct option is :-

(A) $\text{P} \rightarrow 3,4; \text{Q} \rightarrow 1,4; \text{R} \rightarrow 2,4; \text{S} \rightarrow 1,3,4$

(B) $\text{P} \rightarrow 1,2; \text{Q} \rightarrow 2,4; \text{R} \rightarrow 1,2,3; \text{S} \rightarrow 1,3$

(C) $\text{P} \rightarrow 1,2,3; \text{Q} \rightarrow 1,4; \text{R} \rightarrow 1,3; \text{S} \rightarrow 1,2,3$

(D) $\text{P} \rightarrow 3,4; \text{Q} \rightarrow 1,2,3; \text{R} \rightarrow 2,3,4; \text{S} \rightarrow 2,3,4$

Space for Rough Work

9.

List-I (Radicals)		List-II (Correct statement regarding them)	
(P)	Br^-	(1)	Can be confirmed by Brown ring test
(Q)	I^-	(2)	Can be confirmed by Layer test
(R)	S^{2-}	(3)	Can be confirmed by methylene blue test
(S)	NO_3^-	(4)	On reaction with conc. H_2SO_4 , oxidation takes place
		(5)	All salts are soluble

The **CORRECT** option is :-

- (A) $\text{P} \rightarrow 2,4; \text{Q} \rightarrow 2,4; \text{R} \rightarrow 3,5; \text{S} \rightarrow 1,5$ (B) $\text{P} \rightarrow 1,4; \text{Q} \rightarrow 2,3; \text{R} \rightarrow 3,5; \text{S} \rightarrow 1,5$
 (C) $\text{P} \rightarrow 2,4; \text{Q} \rightarrow 2,4; \text{R} \rightarrow 3,4; \text{S} \rightarrow 1,5$ (D) $\text{P} \rightarrow 2,4; \text{Q} \rightarrow 1,4; \text{R} \rightarrow 3,4; \text{S} \rightarrow 2,5$

10.

List-I		List-II	
(P)	Acetophenone	(1)	Reaction with $\text{AgNO}_3 + \text{NH}_4\text{OH}$ to give white ppt
(Q)	Benzaldehyde	(2)	Reaction with NaOI to give yellow ppt of CHI_3
(R)	Butanal	(3)	Reaction with Fehling to give brown ppt of Cu_2O
(S)	1-butyne	(4)	Reaction with $\text{Ac}_2\text{O} + \text{AcONa}$ to give Cinnamic acid

Select correct code for your answer :

- (A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 4; \text{R} \rightarrow 3; \text{S} \rightarrow 1$ (B) $\text{P} \rightarrow 1; \text{Q} \rightarrow 4; \text{R} \rightarrow 3; \text{S} \rightarrow 2$
 (C) $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 1$ (D) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

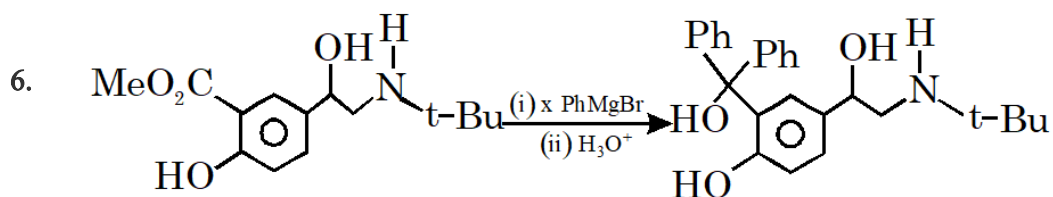
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. During electrolysis, 40 mA current is passed through 100ml of 0.2M $\text{Fe}_2(\text{SO}_4)_3$ solution for certain time. If 30 ml of 0.01M KMnO_4 is decolourised by final solution in cell then find time of electrolysis, in hours. (Given $1F = 96000 \text{ C/mole}$)
 2. An organic compound contains carbon, hydrogen and nitrogen. If weight ratio of C, H, N (in same order) is 9 : 1 : 3.5 and molar mass of compound is 216 gm/mole and number of atoms present in 1 molecule of compound are x then calculate value of $\frac{x}{4}$.
 3. 100 ml of 50 mol m^{-3} $\text{Ba}(\text{OH})_2(\text{aq.})$ solution is mixed with 400 ml 50 mol m^{-3} $\text{HX}(\text{aq.})$ solution. Given : $\text{pK}_a(\text{HX}) = 4$. Find final pH of solution.
 4. How many of the following is/are yellow and water insoluble substances ?
 Ag_2CrO_4 , PbCrO_4 , Na_2CrO_4 ,
 BaCrO_4 , CrO_2Cl_2 , $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$,
 $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$,
 $\text{K}_3[\text{Co}(\text{NO}_2)_6]$, AgI , PbI_2
-

Space for Rough Work

5. How many gases/vapours are involved either as reactants or products in Serpeck's process used for concentration of bauxite.



Number of equivalents of Grignard reagent (x) used in the reaction is

7. Consider all possible isomeric ketones including stereoisomers of MW = 100, All these isomers are independently reacted with NaBH_4 (NOTE : stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are.
8. 2-Butanol $\xrightarrow[\Delta]{\text{H}^+}$ Products If all products formed in the above reaction undergo reaction with $\text{Br}_2(\text{CCl}_4)$ individually then number of products formed will be :

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $a^6 - 6a^5 + 12a^4 + ba^3 + 12a^2 - 6a + 1 \geq 0$ holds for all real a , then all values of the real number b is given by $[p, q]$, then $(q - p + 12)$ is divisible by
(A) 20 (B) 40 (C) 31 (D) 62
2. If $I(a, b) = \int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx$, where $a, b > 0$, then
(A) $I(2, 4) - I(4, 8) = 0$ (B) $I(3, 6) - I(6, 12) = 0$
(C) $\frac{I(4, 16)}{I(2, 4)} = 1$ (D) $\frac{I(9, 81)}{I(3, 9)} = 2$
3. Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined as $f(x) = h(x) + 9x - 4$, where $h(x)$ is some integrable function. If $\int_0^1 f(x)dx = \int_0^1 x f(x)dx = \frac{1}{2}$, then the possible value(s) of $\int_0^1 ((f(x))^2 + (x^2 - x)h'(x)) dx$ can be
(A) $\frac{5}{2}$ (B) $\frac{1}{2}$ (C) 3 (D) 2

Space for Rough Work

4. If $f(x) = \begin{cases} \cos^{-1}x & , x < 0 \\ \sin^{-1}x & , x \geq 0 \end{cases}$, $g(x) = \begin{cases} \sin^{-1}x & , x < 0 \\ \cos^{-1}x & , x \geq 0 \end{cases}$

And $h(x) = \min\{f(x), g(x)\}$ then which of the following is/are true ?

(A) The number of non-differentiable points of $h(x)$ in $(-1, 1)$ is one

(B) Local maximum value of $h(x)$ is $\frac{\pi}{4}$

(C) The area bounded by $y = h(x)$ & x-axis is $(\sqrt{2} - 1)$ sq. units

(D) $h(x)$ is concave up in $\left(0, \frac{1}{\sqrt{2}}\right)$

5. Let $f(x) = \begin{cases} \frac{ax^3+bx^2+cx+d}{x} & \forall x \neq 0 \\ 2 & x = 0 \end{cases}$ is a continuous function where a, b, c, d are in Arithmetic

progression. If $x = p$ be the point of local minimum of $f(x)$ then,

(where $[.]$ represents greatest integer function and $\{.\}$ represents fractional part function)

(A) $[p] = -1$

(B) $\{p\} = \frac{2}{3}$

(C) minimum value of $f(x)$ is $\frac{4}{3}$

(D) $\{p\} = \frac{1}{3}$

6. Let $f(x) = x^{2018} + a_0x^{2017} + a_1x^{2016} + a_2x^{2015} + \dots + a_{2017}$

where $a_0, a_1, a_2, \dots, a_{2017}$ are real numbers such that $f(0) = f(1)$ and each root of $f(x) = 0$ is real and always lies between 0 and 1 then product of roots is

(A) less than or equal to $\frac{1}{2^{2018}}$

(B) less than or equal to $\frac{1}{2^{2017}}$

(C) greater than or equal to $\frac{1}{2^{2018}}$

(D) less than or equal to $\frac{1}{2^{2016}}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Consider $f(x) = x^3 - x + 1$, $x \in \mathbb{R}$,

Let N denotes number of real roots of equation $f(x) = 0$,

M denotes number of local maximum of the function $y = |f(x)|$,

m denotes number of local minima of the function $y = f(|x|)$

and x_0 is abscissa of point of inflection of $y = f(x)$.

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$N + M - x_0$	(1)	1
(Q)	$\frac{N + Mm + x_0}{N}$	(2)	2
(R)	$\frac{(N^{x_0} + m + M)}{(m^{x_0})}$	(3)	3
(S)	$Mm - (N + m + M)^{mx_0}$	(4)	4

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

Space for Rough Work

8. Consider $\int x^3 (\ln x)^2 dx = \frac{x^4}{32} [a(\ln x)^b + c \ln x + d] + k$, (where k is constant of integration and $b > 1$) then match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Value of a is equal to	(1)	1
(Q)	Value of b is equal to	(2)	-4
(R)	Value of c is equal to	(3)	2
(S)	Value of d is equal to	(4)	8

- (A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$ (B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
 (C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$ (D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$
9. Consider a twice differentiable function $f(x)$ satisfying $f(x) + f''(x) = 2f'(x)$ where $f(0) = 0$, $f(1) = e$.
 [Note : e denotes napier's constant and $e = 2.71$]

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$f'(-1)$ is equal to	(1)	0
(Q)	$f''(1)$ is equal to	(2)	1
(R)	$\int_0^1 f(x) dx$ is equal to	(3)	$2e^2$
(S)	Number of roots of the equation $f(x) = 0$ is equal to	(4)	$3e$

- (A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$ (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 2$
 (C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$ (D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Value of $\int_0^1 \frac{dx}{\sqrt{1-x^2} (1 + \sqrt{1-x^2})}$ is equal to	(1)	2
(Q)	Value of $\lim_{x \rightarrow 0} \frac{\int_0^x x e^{t^2} dt}{\left(e^{\frac{x^2}{2}} - 1\right)}$	(2)	4
(R)	$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{2\sqrt{n}}\right)$ equal to	(3)	12
(S)	If $f(x) = \int_{x-1}^{2x} (t^3 - 2t + 4) dt$, then value of $f'(1)$ is equal to	(4)	1

(A) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If the number of solution(s) of the equation, $e^{\int_0^{\infty} \frac{t \, dx}{(x+x^{2021})\left(x+\frac{1}{x}\right)}} = \int_0^t x e^{t^2-x^2} dx$ is λ , then $\frac{\lambda}{6}$ equals
2. If $I_m = \int_0^{\infty} e^{-x} \sin^m x dx$ where $m \geq 2$, $m \in \mathbb{N}$ then $\frac{85I_4}{48}$ equals
3. Value of 'k' so that $f(k) = \int_0^4 |4x - x^2 - k| dx$ is minimum is
4. Let $4 \left(\frac{ay}{\cos \alpha} - \frac{bx}{\sin \alpha} \right) = 3(a+b)$, $ab \neq 0$ is normal to the curve $x = 3\cos\theta - \cos^3\theta$, $y = 3\sin\theta - \sin^3\theta$ at $\theta = \alpha$, then number of possible values of α , $\alpha \in (0, \pi) - \left\{ \frac{\pi}{2} \right\}$ is

Space for Rough Work

5. If the solution of the differential equation $\frac{dy}{dx} - y \left(\frac{x^4 + 3x^2}{(x^2 + 1)^2} \right) = (4x + 3)e^{\frac{x^3}{x^2 + 1}}$, is in the form $y = f(x)$ (where $f(0) = 1$), then $\frac{f(1) + f(-1)}{4\sqrt{e}}$ is
6. Let $f(x) = |x - 1| + |2x - 1| + |3x - 1| + \dots + |119x - 1|$. If 'm' denotes the minimum value of $f(x)$ then $\frac{m}{10}$ equals
7. If $f(n)$ be defined on the set of natural numbers such that $f(1) + 2^3f(2) + 3^3f(3) + \dots + n^3f(n) = n^4f(n)$, $\forall n \in \mathbb{N}$, where $f(1) = (2018)^3$, then $\frac{1}{35} \left(\frac{f(1009) - 1}{2} + \int_{f(2018)}^{f(1009)} ([x] - \{x\}) dx \right)$ (where $[.]$ and $\{.\}$ are G.I.F and fractional part functions respectively) equals
8. Value of $\frac{29 \int_0^1 (1 - x^4)^7 dx}{4 \int_0^1 (1 - x^4)^6 dx}$ is equal to

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work